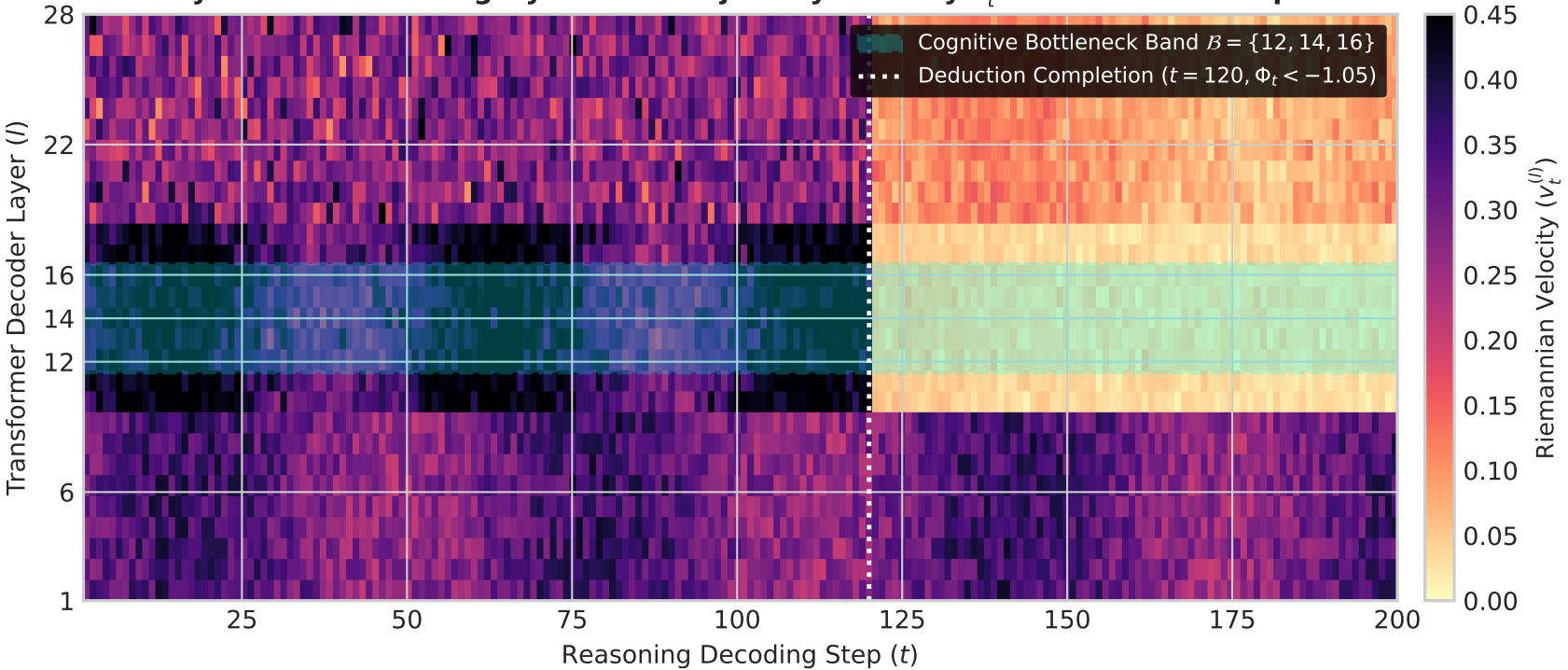

Layer-wise Reasoning Dynamics: Trajectory Velocity $v_t^{(l)}$ across Model Depth
Cognitive Bottleneck Band $\mathcal{B} = \{12, 14, 16\}$
Deduction Completion ($t = 120, \Phi_t < -1.05$)
Transformer Decoder Layer (l)
28
22
16
14
12
6
1
25
50
75
100
125
150
175
200
Reasoning Decoding Step (t)
Riemannian Velocity ($v_t^{(l)}$)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00